TABLE 54 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	3.58	4.21	4.93	5.77	6.77	7.96	9.38	11.09	17.6%
Japan	1.67	1.97	2.32	2.73	3.21	3.80	4.50	5.34	18.2%
India	0.21	0.25	0.29	0.34	0.40	0.46	0.55	0.64	17.3%
Rest of Asia Pacific	3.01	3.51	4.10	4.78	5.58	6.53	7.67	9.03	17.1%
Total	**8.47**	**9.93**	**11.63**	**13.62**	**15.96**	**18.75**	**22.09**	**26.12**	**17.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 55 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.91	1.01	1.12	1.24	1.38	1.53	1.71	1.92	11.4%
Rest of Latin America	1.02	1.13	1.26	1.39	1.54	1.71	1.91	2.13	11.2%
Total	**1.93**	**2.14**	**2.37**	**2.63**	**2.92**	**3.25**	**3.62**	**4.05**	**11.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

8.3 SEQUENCING BY LIGATION (SBL)

8.3.1 INCREASING DEMAND FOR SBL IN FOOD MICROBIOME RESEARCH TO BOOST GROWTH

SBL is a DNA sequencing method that uses DNA ligase to identify the nucleotide present at a given position in a DNA sequence. SBL analyzes the mismatch in base pairs of a DNA ligase to determine the underlying sequence of a target DNA molecule.

SOLiD is an enzymatic method of sequencing that uses DNA ligase, an enzyme used widely in biotechnology, for its ability to ligate double-stranded DNA strands. Emulsion PCR is used to immobilize/amplify an ssDNA primer-binding region (known as an adapter), which has been conjugated to the target sequence on a bead. These beads are then deposited onto a glass surface with a high density of beads. This, in turn, increases the throughput of the technique.

TABLE 56 MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	16.07	18.15	20.49	23.15	26.18	29.71	33.83	38.69	13.6%
Europe	8.75	9.96	11.32	12.89	14.69	16.80	19.27	22.21	14.4%
Asia Pacific	2.42	2.83	3.29	3.84	4.48	5.23	6.13	7.22	17.0%
Latin America	0.55	0.61	0.68	0.75	0.82	0.91	1.01	1.13	10.8%
Middle East and Africa	0.38	0.42	0.46	0.50	0.54	0.59	0.65	0.71	9.2%
Total	**28.18**	**31.96**	**36.24**	**41.13**	**46.71**	**53.24**	**60.89**	**69.95**	**14.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 57 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	14.56	16.46	18.60	21.03	23.79	27.02	30.79	35.23	13.6%
Canada	1.51	1.69	1.89	2.13	2.39	2.69	3.04	3.45	12.8%
Total	**16.07**	**18.15**	**20.49**	**23.15**	**26.18**	**29.71**	**33.83**	**38.69**	**13.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 58 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	1.89	2.16	2.46	2.80	3.20	3.67	4.22	4.87	14.7%
UK	1.69	1.94	2.22	2.55	2.93	3.37	3.90	4.52	15.3%
France	1.34	1.52	1.73	1.98	2.26	2.58	2.97	3.43	14.6%
Italy	0.38	0.43	0.49	0.55	0.63	0.72	0.83	0.95	14.3%
Spain	0.29	0.33	0.37	0.42	0.48	0.55	0.62	0.72	14.0%
Rest of Europe	3.16	3.58	4.05	4.59	5.19	5.91	6.74	7.72	13.8%
Total	**8.75**	**9.96**	**11.32**	**12.89**	**14.69**	**16.80**	**19.27**	**22.21**	**14.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 59 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	1.02	1.19	1.39	1.62	1.89	2.21	2.58	3.04	16.9%
Japan	0.48	0.56	0.66	0.77	0.90	1.06	1.25	1.47	17.6%
India	0.06	0.07	0.08	0.10	0.11	0.13	0.15	0.18	16.8%
Rest of Asia Pacific	0.86	1.00	1.17	1.36	1.58	1.84	2.15	2.53	16.7%
Total	**2.42**	**2.83**	**3.29**	**3.84**	**4.48**	**5.23**	**6.13**	**7.22**	**17.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 60 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.26	0.29	0.32	0.35	0.39	0.43	0.48	0.53	10.8%
Rest of Latin America	0.29	0.32	0.36	0.39	0.44	0.48	0.53	0.60	10.7%
Total	**0.55**	**0.61**	**0.68**	**0.75**	**0.82**	**0.91**	**1.01**	**1.13**	**10.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

8.4 NANOPORE SEQUENCING

8.4.1 ONLY SEQUENCING TECHNOLOGY WITH DIRECT RNA SEQUENCING CAPABILITIES

Introduced by Oxford Nanopore Technologies, nanopore sequencing technology (which is considered the fourth generation of sequencing) has emerged as a competitive, portable technology.

The technology works through the detection of unique electrical signals of different molecules as they pass across the nanopore with a semiconductor-based electronic detection system. This technology makes for a high throughput, cost-effective sequencing solution. The technology's core/heart consists of the biological nanopore, a protein pore embedded in a membrane, while the brains of the technology lie in the electronics of a semiconductor integrated circuit and proprietary chemistries. The electronic sensor embedded in the chip enables automatic membrane assembly and nanopore insertion while allowing for active control of individual sensors on the circuit. Various types of sequencing chemistries can be paired with nanopore and electronic sensor technology, which enables high-throughput and high-accuracy sequencing. However, the low accuracy of nanopore sequencing as compared to other well-established techniques is one of the major factors restraining the growth of this market segment.

In mid-2014, Oxford Nanopore Technologies launched MinION, which was the only nanopore sequencer commercially available in the market until 2015. This sequencer is portable, weighs just 100 gm, and can sequence DNA and RNA of up to ~200kb in length. Due to its portable nature, the MinION sequencer can